Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Made with GAMMA

Dataset Summary

3,900

Rows

Total purchases analyzed

18

Columns

Data fields captured

37

Missing Values

In Review Rating column

Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Made with GAMMA

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

01

Data Loading

Imported the dataset using `pandas`.

02

Initial Exploration

Used `df.info()` to check structure and `.describe()` for summary statistics.

03

Missing Data Handling

Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.

04

Column Standardization

Rename columns to **snake case** for better readability and documentation.

05

Feature Engineering

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

06

Data Consistency Check

Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

07

Database Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Made with GAMMA

Statistical Summary

Customer Demographics & Purchase Details

Customer ID	count: 3900.000000	mean: 1950.500000
Age	count: 3900.000000	mean: 44.068462
Gender	unique: 2	top: Male
Item Purchased	unique: 25	top: Blouse
Category	unique: 4	top: Clothing
Purchase Amount (USD)	count: 3900.000000	mean: 59.764359
Location	unique: 50	top: Montana
Size	unique: 4	top: M
Color	unique: 25	top: Olive
Season	unique: 4	top: Spring
Review Rating	count: 3863.000000	mean: 3.750065
Subscription Status	unique: 2	top: No
Shipping Type	unique: 6	top: Free Shipping
Discount Applied	unique: 2	top: No

Payment & Purchase Behavior

Discount Applied	unique: 2	top: No
Promo Code Used	unique: 2	top: No
Previous Purchases	count: 3900.000000	mean: 25.351538
Payment Method	unique: 6	top: PayPal
Frequency of Purchases	unique: 7	top: Every 3 Months

Made with GAMMA

SQL Analysis Results: Revenue & Ratings

Revenue by Gender

Compared total revenue generated by male vs. female customers.

gender	revenue
Female	75191
Male	157890

Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Made with GAMMA

SQL Analysis Results: Shipping & Subscriptions

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	round
Standard	58.46
Express	60.48

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

Made with GAMMA

SQL Analysis Results: Products & Segments

Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
New	83
Returning	701

Made with GAMMA

SQL Analysis Results: Products & Repeat Buyers

Top 3 Products per Category

Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2518
Yes	958

Made with GAMMA

SQL Analysis Results: Revenue by Age

Calculated total revenue contribution of each age group.

27%

Young Adult

$62,143 revenue

25%

Middle-aged

$59,197 revenue

24%

Adult

$55,978 revenue

24%

Senior

$55,763 revenue

Made with GAMMA

Dashboard & Business Recommendations

Customer Behavior Dashboard

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

KPIs:

- **3.9K** Number of Customers
- **$59.76** Average Purchase Amount
- **3.75** Average Review Rating

Filters:

- **Subscription Status:** No, Yes
- **Gender:** Female, Male
- **Category:** Accessories, Clothing, Footwear, Outerwear
- **Shipping Type:** 2-Day Shipping, Express, Free Shipping, Next Day Air, Standard, Store Pickup

Visualizations:

- **% of Customers by Subscription Status:** Donut chart showing 27% Yes and 73% No.
- **Revenue by Category:** Bar chart showing revenue for Clothing, Accessories, Footwear, and Outerwear.
- **Sales by Category:** Bar chart showing sales for Clothing, Accessories, Footwear, and Outerwear.
- **Revenue by Age Group:** Horizontal bar chart showing revenue for Young Adult, Middle-aged, Adult, and Senior.
- **Sales by Age Group:** Horizontal bar chart showing sales for Young Adult, Middle-aged, Senior, and Adult.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

Made with GAMMA